Signals and Systems Lecture-8: Fourier Series (2)

Chadi Abou-Rjeily

Department of Electrical and Computer Engineering
Lebanese American University
chadi.abourjeily@lau.edu.lb

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Calculating the Fourier series: Example-1 (1)

Example: Determine the exponential Fourier coefficients of:

$$x(t) = \frac{1}{2} - \frac{3}{4}\cos(20\pi t) + \frac{1}{2}\sin(30\pi t)$$

- Determine the period of $x(t)$:
  - $\cos(20\pi t) = \cos(2\pi f_1 t)$ with $f_1 = 10$ Hz ($T_1 = \frac{1}{f_1} = \frac{1}{10}$ s).
  - $\sin(30\pi t) = \sin(2\pi f_2 t)$ with $f_2 = 15$ Hz ($T_2 = \frac{1}{f_2} = \frac{1}{15}$ s).
  - The greatest common divisor between f_1 and f_2 is equal to 5. Therefore, the fundamental frequency of $x(t)$ is $f_0 = 5$ Hz and its period is $T_0 = \frac{1}{5}$ s.
  - During each period T_0, $\cos(20\pi t)$ spans 2 periods T_1 while $\sin(30\pi t)$ spans 3 periods T_2.
- In terms of the fundamental frequency f_0, $x(t)$ can be written as:

$$x(t) = \frac{1}{2} - \frac{3}{4}\cos(2\pi(2f_0)t) + \frac{1}{2}\sin(2\pi(3f_0)t)$$

Calculating the Fourier series: Example-1 (2)

First Method: Compare $x(t)$ with the trigonometric FS representation:
$X_c[0] + \sum_{k=1}^{+\infty} X_c[k]\cos(2\pi(kf_0)t + \sum_{k=1}^{+\infty} X_s[k]\sin(2\pi(kf_0)t$

$$x(t) = \underbrace{\frac{1}{2}}_{=X_c[0]} \underbrace{-\frac{3}{4}}_{=X_c[2]} \cos(2\pi(2f_0)t) + \underbrace{\frac{1}{2}}_{=X_s[3]} \sin(2\pi(3f_0)t)$$

- Therefore:
  - $X_c[0] = \frac{1}{2}$.
  - $X_c[2] = -\frac{3}{4}$.
  - $X_c[k] = 0$ for $k \neq 0, 2$.
  - $X_s[3] = \frac{1}{2}$.
  - $X_s[k] = 0$ for $k \neq 3$.

- Consequently:
  - $X[0] = X_c[0] = \frac{1}{2}$.
  - $X[2] = \frac{X_c[2] - jX_s[2]}{2} = -\frac{3}{8}$.
  - $X[3] = \frac{X_c[3] - jX_s[3]}{2} = -j\frac{1}{4}$.
  - $X[-2] = X^*[2] = -\frac{3}{8}$.
  - $X[-3] = X^*[3] = j\frac{1}{4}$.
  - $X[k] = 0$ for $k \neq 0, \pm 2, \pm 3$.

Calculating the Fourier series: Example-1 (3)

Second Method:

- $\cos\theta$ and $\sin\theta$ can be written as:

$$\cos\theta = \frac{e^{j\theta} + e^{-j\theta}}{2} \quad ; \quad \sin\theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$$

- Consequently:

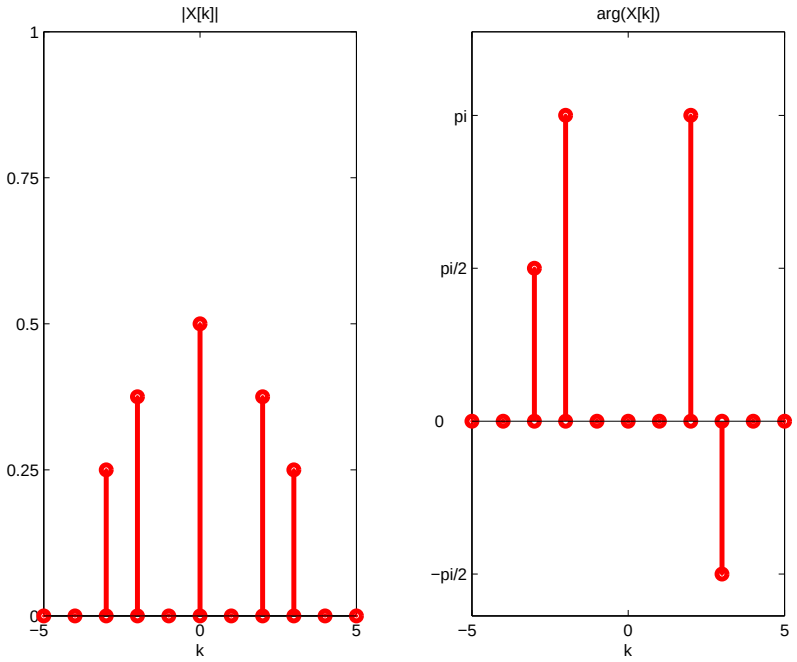
$$x(t) = \frac{1}{2} - \frac{3}{4}\cos(2\pi(2f_0)t) + \frac{1}{2}\sin(2\pi(3f_0)t)$$

$$= \underbrace{\frac{1}{2}}_{=X[0]} \underbrace{-\frac{3}{8}}_{=X[2]} e^{j2\pi(2f_0)t} \underbrace{-\frac{3}{8}}_{=X[-2]} e^{-j2\pi(2f_0)t} + \underbrace{\frac{1}{4j}}_{=X[3]} e^{j2\pi(3f_0)t} \underbrace{-\frac{1}{4j}}_{=X[-3]} e^{-j2\pi(3f_0)t}$$

- $X[0] = \frac{1}{2}$.
- $X[2] = -\frac{3}{8}$.
- $X[3] = -j\frac{1}{4}$.
- $X[-2] = -\frac{3}{8}$.
- $X[-3] = j\frac{1}{4}$.
- $X[k] = 0 \; k \neq 0, \pm 2, \pm 3$.

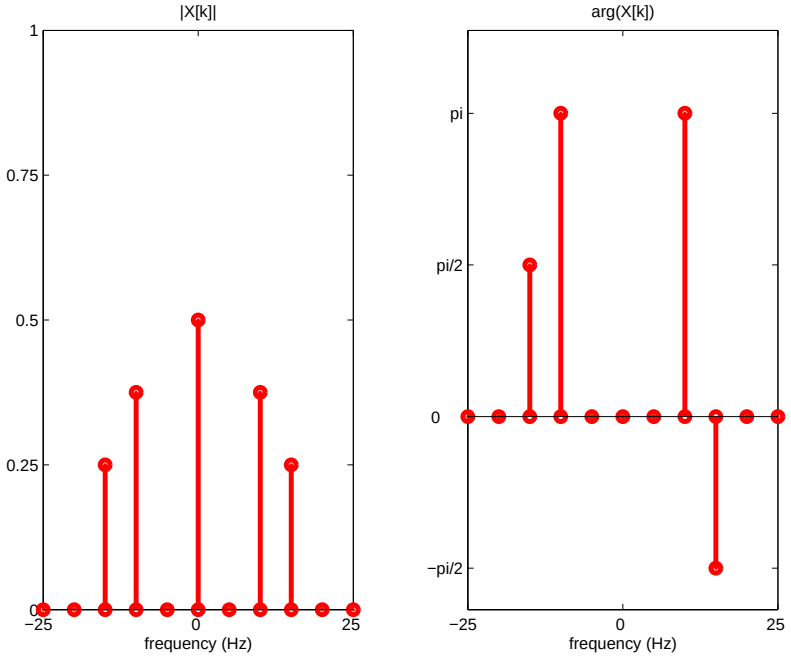
Calculating the Fourier series: Example-1 (4)

- The FS coefficients are characterized by:
  - $X[0] = \frac{1}{2} + 0.j \Rightarrow |X[0]| = \frac{1}{2}$ and $\arg(X[0]) = 0$.
  - $X[-2] = -\frac{3}{8} + 0.j \Rightarrow |X[-2]| = \frac{3}{8}$ and $\arg(X[-2]) = \pi$.
  - $X[2] = -\frac{3}{8} + 0.j \Rightarrow |X[2]| = \frac{3}{8}$ and $\arg(X[2]) = \pi$.
  - $X[-3] = 0 + j\frac{1}{4} \Rightarrow |X[-3]| = \frac{1}{4}$ and $\arg(X[-3]) = \frac{\pi}{2}$.
  - $X[3] = 0 - j\frac{1}{4} \Rightarrow |X[3]| = \frac{1}{4}$ and $\arg(X[3]) = -\frac{\pi}{2}$.
- The magnitudes and the phases of the FS coefficients can be plotted as a function of the harmonic number k:



Calculating the Fourier series: Example-1 (5)

- Given that the k-th harmonic corresponds to the frequency kf_0, then the magnitudes and the phases of the FS coefficients can be plotted as a function of the frequency. In this example, $f_0 = 5$ Hz and:



- These will be referred to as the **magnitude spectrum** (for $|X[k]|$) and **phase spectrum** (for $\arg(X[k])$), respectively.

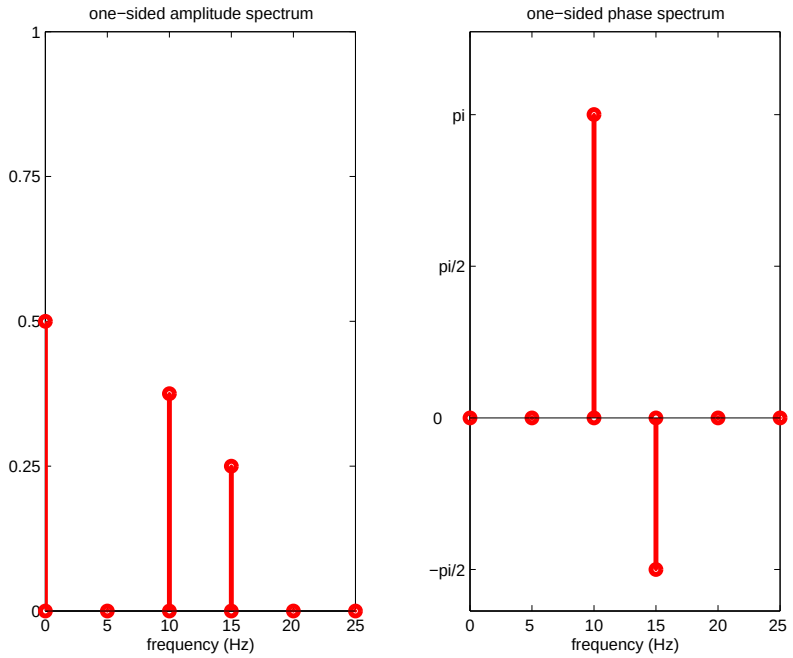
Calculating the Fourier series: Example-1 (6)

- Given that for real signals, $X[-k]$ is related to $X[k]$ by: $X[-k] = X^*[k]$ then:

$$|X[-k]| = |X[k]|$$
$$\arg(X[-k]) = -\arg(X[k])$$

- Consequently, for real signals:
  - The magnitude spectrum is even.
  - The phase spectrum is odd.

- Therefore, in some cases, we plot the magnitude and phase spectra only for positive frequencies. These are referred to as the **one-sided spectra**.



Calculating the Fourier series: Example-1 (7)

- Applying Parseval's theorem, the average power of $x(t)$ can be calculated from:

$$P_{av} = \sum_{k=-\infty}^{+\infty} |X[k]|^2 = |X[0]|^2 + 2|X[2]|^2 + 2|X[3]|^2$$

$$= \frac{1}{4} + 2\left(\frac{3}{8}\right)^2 + 2\left(\frac{1}{4}\right)^2 = \frac{21}{32} \text{ W}$$

- The DC power is given by:

$$P_0 = |X[0]|^2 = \left(\frac{1}{2}\right)^2 = \frac{1}{4} \text{ W}$$

- The power contained in the third harmonic is given by:

$$P_3 = 2|X[3]|^2 = 2\left(\frac{1}{4}\right)^2 = \frac{1}{8} \text{ W}$$

- The average value of the signal is:

$$x_{\text{av}} = X[0] = 1/2$$

Calculating the Fourier series: Example-1 (8)

- Note also that it was possible to calculate the average power without applying Parseval's theorem:

$$P_{av} = \frac{1}{T_0}\int_{T_0} |x(t)|^2 \mathrm{d}t = \frac{1}{T_0}\int_{T_0} \left[\frac{1}{2} - \frac{3}{4}\cos(2\pi(2f_0)t) + \frac{1}{2}\sin(2\pi(3f_0)t)\right]^2 \mathrm{d}t$$

$$= \left(\frac{1}{2}\right)^2 \underbrace{\frac{1}{T_0}\int_{T_0} 1\mathrm{d}t}_{=1} + \left(\frac{3}{4}\right)^2 \underbrace{\frac{1}{T_0}\int_{T_0} \cos^2(2\pi(2f_0)t)\mathrm{d}t}_{=1/2} + \left(\frac{1}{2}\right)^2 \underbrace{\frac{1}{T_0}\int_{T_0} \sin^2(2\pi(3f_0)t)\mathrm{d}t}_{=1/2}$$

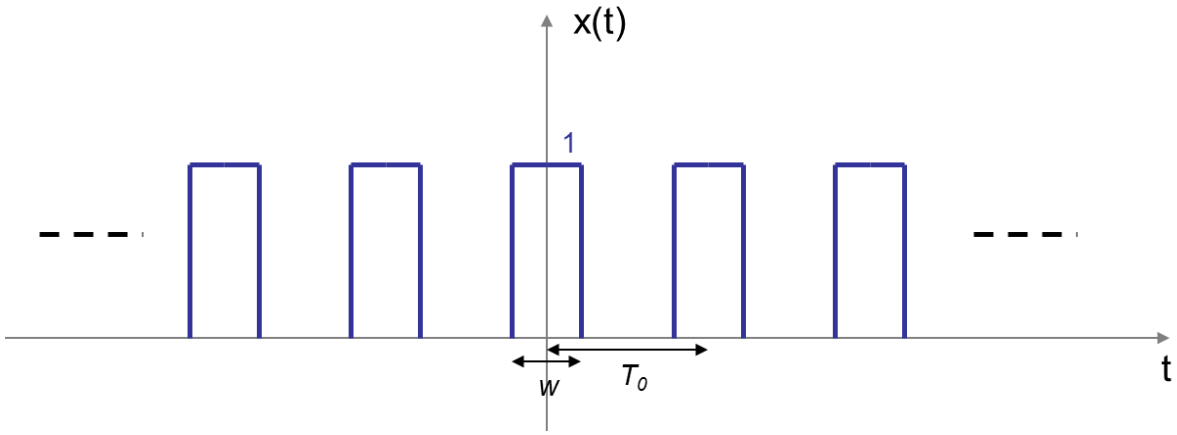
$$- 2\frac{1}{2}\frac{3}{4}\underbrace{\frac{1}{T_0}\int_{T_0} \cos(2\pi(2f_0)t)\mathrm{d}t}_{=0} + 2\frac{1}{2}\frac{1}{2}\underbrace{\frac{1}{T_0}\int_{T_0} \sin(2\pi(3f_0)t)\mathrm{d}t}_{=0}$$

$$- 2\frac{3}{4}\frac{1}{2}\underbrace{\frac{1}{T_0}\int_{T_0} \cos(2\pi(2f_0)t)\sin(2\pi(3f_0)t)\mathrm{d}t}_{=0} = \frac{1}{4} + \frac{9}{32} + \frac{1}{8} = \frac{21}{32} \ \mathrm{W}$$

- Note that the above equalities follow from the orthogonality properties of the sinusoidal functions.

Calculating the Fourier series: Example-2 (1)

Consider the periodic signal obtained from repeating the signal $\text{rect}\left(\frac{t}{w}\right)$ with a period T_0.



- This signal can be represented mathematically by:

$$x(t) = \sum_{n\in\mathbb{Z}} \text{rect}\left(\frac{t-nT_0}{w}\right) = \sum_{n\in\mathbb{Z}} \text{rect}\left(\frac{t}{w}\right) * \delta(t-nT_0)$$

$$= \text{rect}\left(\frac{t}{w}\right) * \sum_{n\in\mathbb{Z}} \delta(t-nT_0) = \text{rect}\left(\frac{t}{w}\right) * \frac{1}{T_0}\text{comb}\left(\frac{t}{T_0}\right)$$

  - In fact: $\text{comb}(at) = \frac{1}{a}\sum \delta(t-\frac{n}{a})$. For $a = \frac{1}{T_0}$, $\text{comb}(\frac{t}{T_0}) = T_0 \sum \delta(t-nT_0) \Rightarrow \frac{1}{T_0}\text{comb}(\frac{t}{T_0}) = \sum \delta(t-nT_0)$.

Calculating the Fourier series: Example-2 (2)

- Given that the function is even, then $X_s[k] = 0$ for all values of k. Therefore, we only need to determine the values of $X_c[0]$ and $X_c[k]$ for $k = 1, 2, \ldots$
- For $k = 0$:

$$X_c[0] = \frac{1}{T_0}\int_{T_0} x(t)\mathrm{d}t = \frac{1}{T_0}\int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} x(t)\mathrm{d}t$$

  - Since $x(t)$ is even:

$$X_c[0] = \frac{2}{T_0}\int_{0}^{\frac{T_0}{2}} x(t)\mathrm{d}t = \frac{2}{T_0}\int_{0}^{\frac{w}{2}} 1.\mathrm{d}t = \frac{w}{T_0}$$

Calculating the Fourier series: Example-2 (3)

- For $k \neq 0$:

$$X_c[k] = \frac{2}{T_0} \int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} x(t) \cos\left(2\pi(kf_0)t\right) \mathrm{d}t$$

  - Since $x(t)$ is even and $\cos\left(2\pi(kf_0)t\right)$ is even, then their product is even and:

$$X_c[k] = \frac{4}{T_0} \int_0^{\frac{T_0}{2}} x(t) \cos\left(2\pi(kf_0)t\right) \mathrm{d}t = \frac{4}{T_0} \int_0^{\frac{w}{2}} \cos\left(2\pi(kf_0)t\right) \mathrm{d}t$$
$$= \frac{4}{T_0} \frac{1}{2\pi k f_0} \left[\sin\left(2\pi(kf_0)t\right)\right]_0^{\frac{w}{2}} = \frac{2}{\pi k} \sin\left(\pi \frac{kw}{T_0}\right)$$

- Finally:

$$X_c[k] = 2\frac{w}{T_0} \frac{1}{\pi k w / T_0} \sin\left(\pi \frac{kw}{T_0}\right) = 2\frac{w}{T_0} \mathrm{sinc}\left(\frac{w}{T_0} k\right)$$

  - since $\mathrm{sinc}(x) = \frac{\sin(\pi x)}{\pi x}$.

Calculating the Fourier series: Example-2 (4)

The exponential FS coefficients can be determined from the trigonometric FS coefficients by:

$$X[0] = X_c[0] = \frac{w}{T_0}$$

$$X[k] = \frac{X_c[k] - jX_s[k]}{2} = \frac{X_c[k]}{2} = \frac{w}{T_0}\text{sinc}\left(\frac{w}{T_0}k\right) \quad ; \quad k = 1, 2, \ldots$$

$$X[-k] = X^*[k] = \frac{w}{T_0}\text{sinc}\left(\frac{w}{T_0}k\right) \quad ; \quad k = 1, 2, \ldots$$

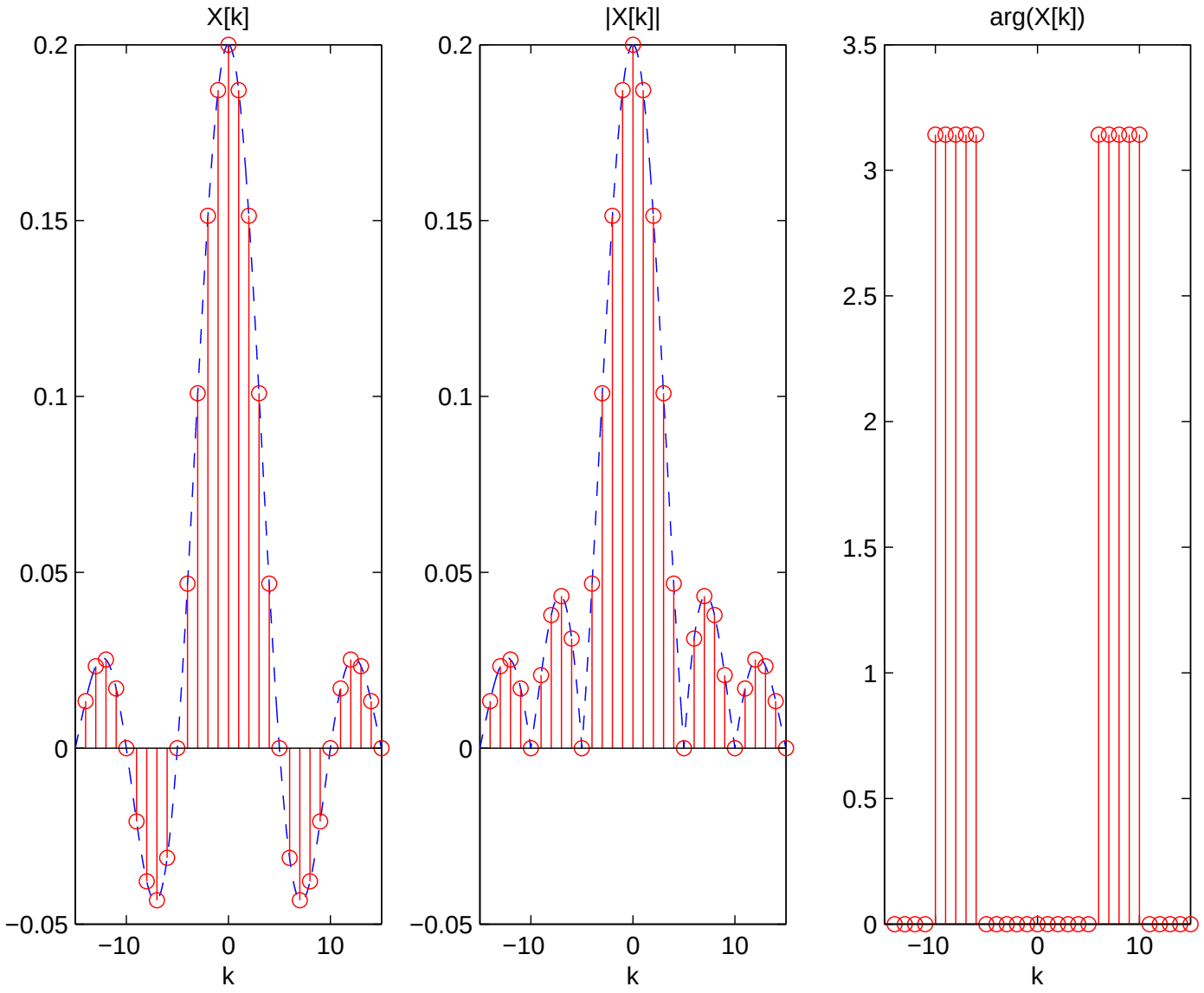
The above relations can be combined into one equation:

$$X[k] = \frac{w}{T_0}\text{sinc}\left(\frac{w}{T_0}k\right) \quad ; \quad k \in \mathbb{Z}$$

- Example, for $\frac{T_0}{w} = 5$, then $X[k] = \frac{1}{5}\text{sinc}\left(\frac{k}{5}\right)$. In this case, $X[k] = 0$ when $\frac{k}{5}$ is a nonzero integer implying that k is a nonzero multiple of 5.

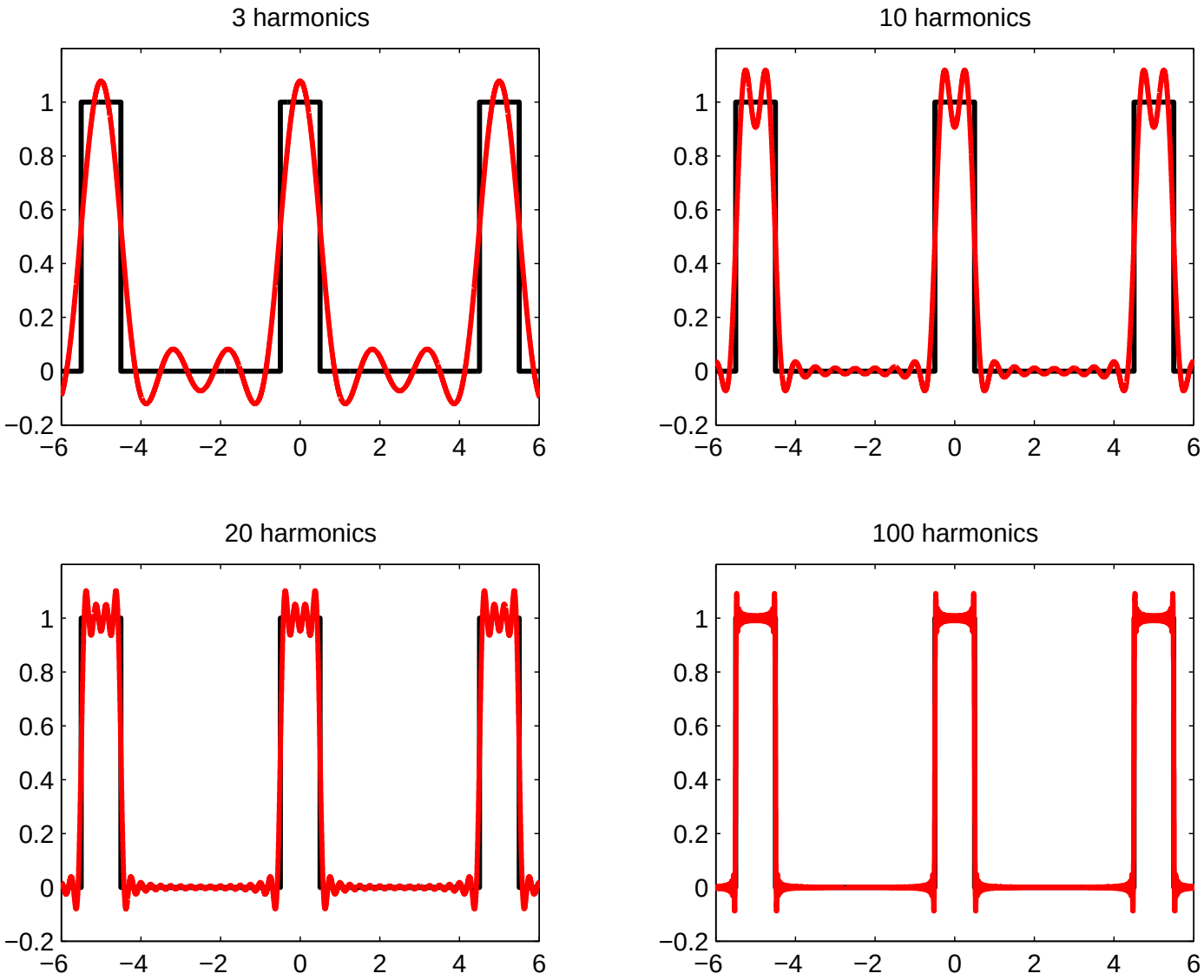
Calculating the Fourier series: Example-2 (5)

- Example: for $\frac{T_0}{w} = 5$:



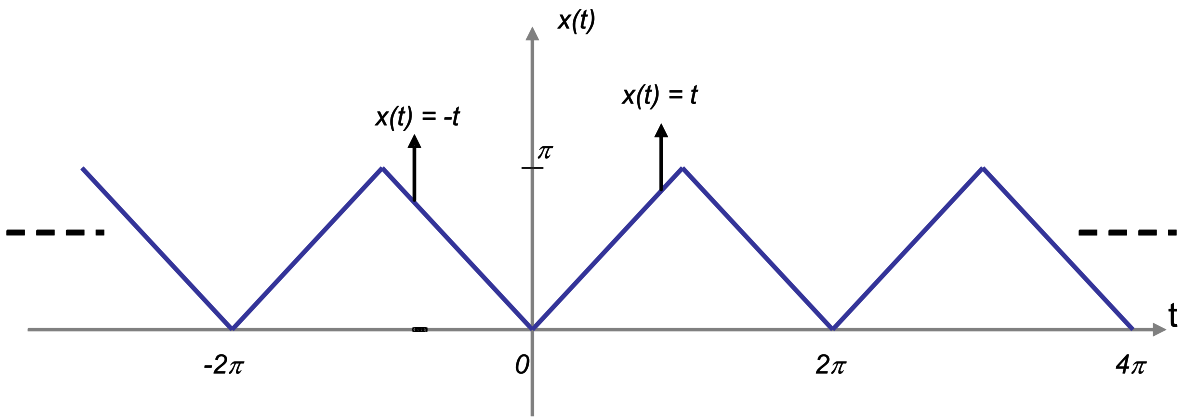
Calculating the Fourier series: Example-2 (6)

- Example: $w = 1$ and $T_0 = 5$:



Calculating the Fourier series: Example-3 (1)

Consider the continuous-time signal given by:



This is a periodic signal with period $T_0 = 2\pi$.

- Since $x(t)$ is an even function, then $X_s[k] = 0 \; \forall \; k$.
- For $k = 0$ and since $x(t)$ is even:

$$X_c[0] = \frac{1}{2\pi}\int_{-\pi}^{\pi} x(t)\mathrm{d}t = \frac{2}{2\pi}\int_{0}^{\pi} x(t)\mathrm{d}t$$

$$= \frac{1}{\pi}\int_{0}^{\pi} t\mathrm{d}t = \frac{1}{\pi}\left[\frac{t^2}{2}\right]_0^{\pi} = \frac{\pi}{2}$$

Calculating the Fourier series: Example-3 (2)

- For $k \neq 0$:

$$X_c[k] = \frac{2}{2\pi} \int_{-\pi}^{+\pi} x(t) \cos\left(2\pi(kf_0)t\right) \mathrm{d}t$$

  - Since $f_0 = \frac{1}{T_0} = \frac{1}{2\pi}$, then:

$$X_c[k] = \frac{1}{\pi} \int_{-\pi}^{+\pi} x(t) \cos\left(kt\right) \mathrm{d}t$$

  - Since $x(t)$ is even and $\cos(kt)$ is even then $x(t)\cos(kt)$ is even and:

$$X_c[k] = \frac{2}{\pi} \int_{0}^{+\pi} x(t) \cos\left(kt\right) \mathrm{d}t$$

Calculating the Fourier series: Example-3 (3)

$$X_c[k] = \frac{2}{\pi}\int_0^{+\pi} t\cos(kt)\,\mathrm{d}t$$

- Now we have to apply the integration by parts:

$$\int u\frac{\mathrm{d}v}{\mathrm{d}t}\mathrm{d}t = uv - \int v\frac{\mathrm{d}u}{\mathrm{d}t}\mathrm{d}t$$

$$u = t \;\Rightarrow\; \frac{\mathrm{d}u}{\mathrm{d}t} = 1$$

$$\frac{\mathrm{d}v}{\mathrm{d}t} = \cos(kt) \;\Rightarrow\; v = \frac{1}{k}\sin(kt)$$

- Therefore:

$$\int t\cos(kt)\mathrm{d}t = \frac{t}{k}\sin(kt) - \int \frac{1}{k}\sin(kt)\mathrm{d}t$$

$$= \frac{t}{k}\sin(kt) + \frac{1}{k^2}\cos(kt)$$

Calculating the Fourier series: Example-3 (4)

- Consequently:

$$X_c[k] = \frac{2}{\pi}\left[\frac{t}{k}\sin(kt) + \frac{1}{k^2}\cos(kt)\right]_0^{\pi}$$
$$= \frac{2}{\pi}\left[\frac{\pi}{k}\sin(k\pi) + \frac{1}{k^2}\left(\cos(k\pi) - 1\right)\right]$$

- Given that $\sin(k\pi) = 0$ and $\cos(k\pi) = (-1)^k$ then:

$$X_c[k] = \frac{2}{\pi k^2}\left((-1)^k - 1\right)$$

- Finally, for $k \neq 0$

$$X_c[k] = \begin{cases} 0, & \text{if } k \text{ is even} \\ -\frac{4}{\pi k^2}, & \text{if } k \text{ is odd} \end{cases}$$

Calculating the Fourier series: Example-3 (5)

- The exponential FS coefficients can be determined from the trigonometric FS coefficients by:

$$X[0] = X_c[0] = \frac{\pi}{2}$$

$$X[k] = \frac{X_c[k] - jX_s[k]}{2} = \frac{X_c[k]}{2} = -\frac{2}{\pi k^2} \quad k = 1, 3, 5, 7 \ldots$$

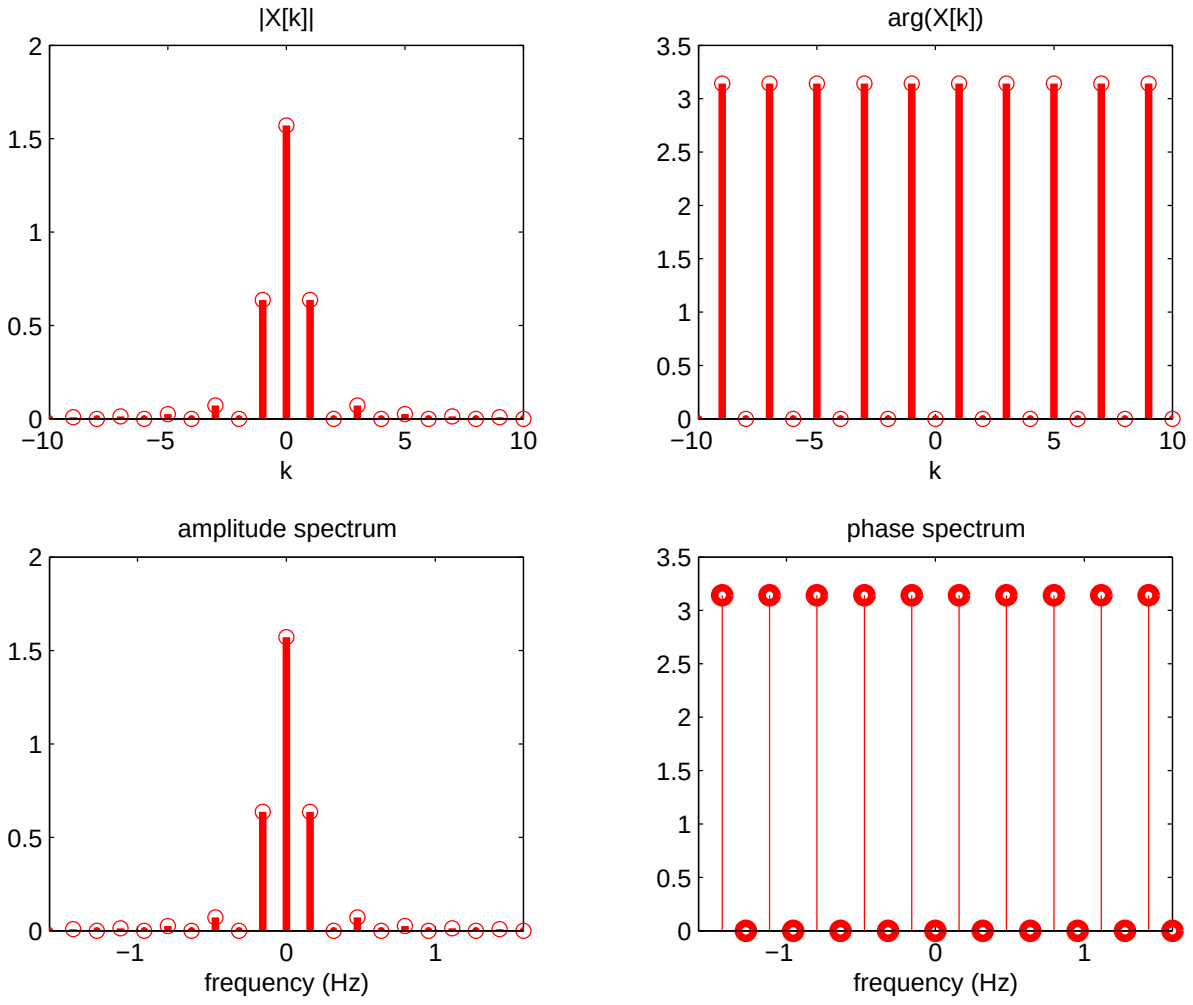
$$X[k] = 0 \quad k = 2, 4, 6, 8 \ldots$$

$$X[-k] = X^*[k] = -\frac{2}{\pi k^2} \quad k = 1, 3, 5, 7 \ldots$$

$$X[-k] = X^*[k] = 0 \quad k = 2, 4, 6, 8 \ldots$$

Calculating the Fourier series: Example-3 (6)

- The magnitude and phase spectra are shown below:



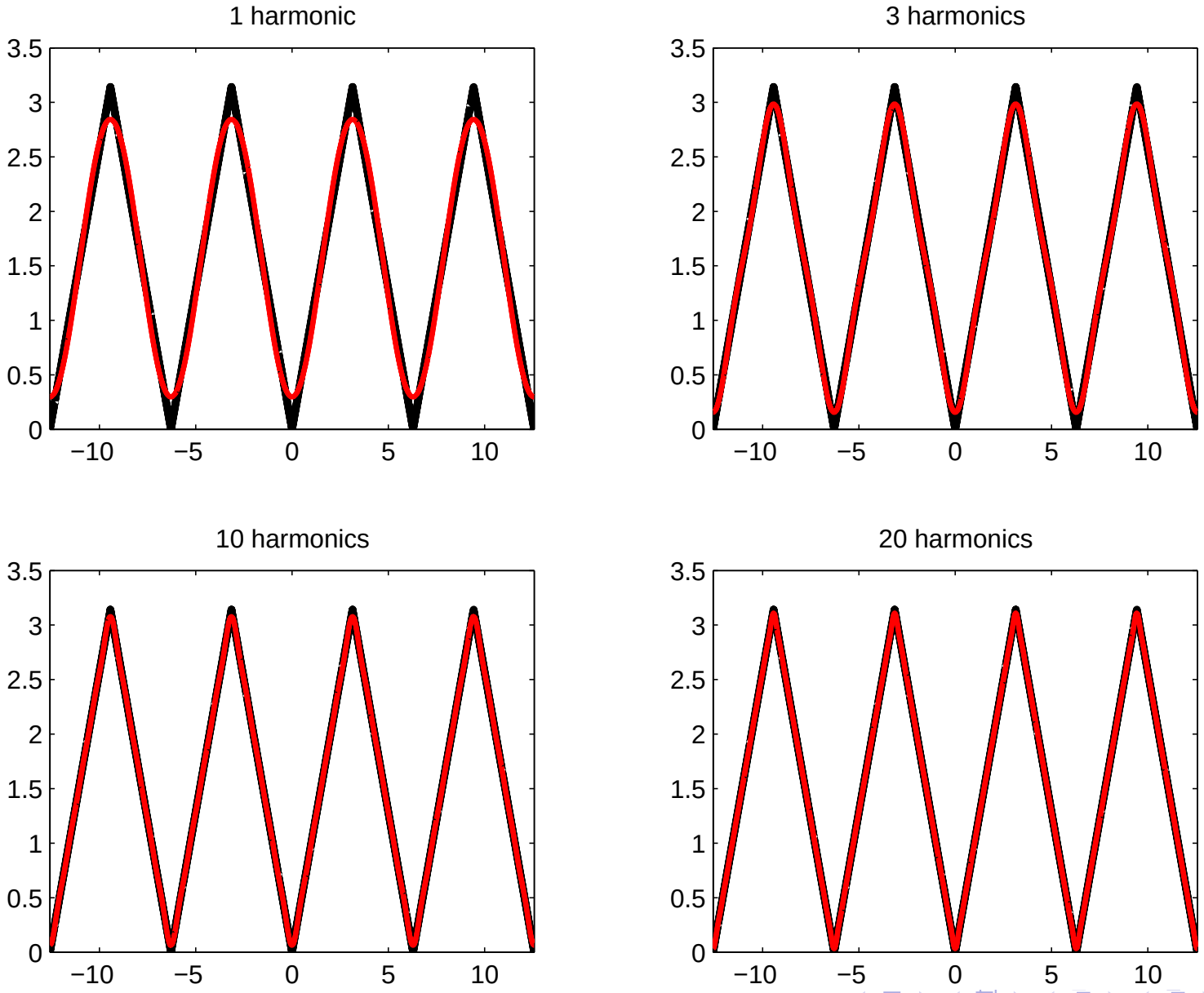
Calculating the Fourier series: Example-3 (7)

- The DC power is:

$$P_0 = |X[0]|^2 = \frac{\pi^2}{4}$$

- The power included in the even harmonics is equal to zero.
- The power included in the k-th odd harmonic is equal to:

$$P_k = 2|X[k]|^2 = \frac{8}{\pi^2 k^4}$$

- Consequently, the main power of the signal power is included in its DC component.

Calculating the Fourier series: Example-3 (8)